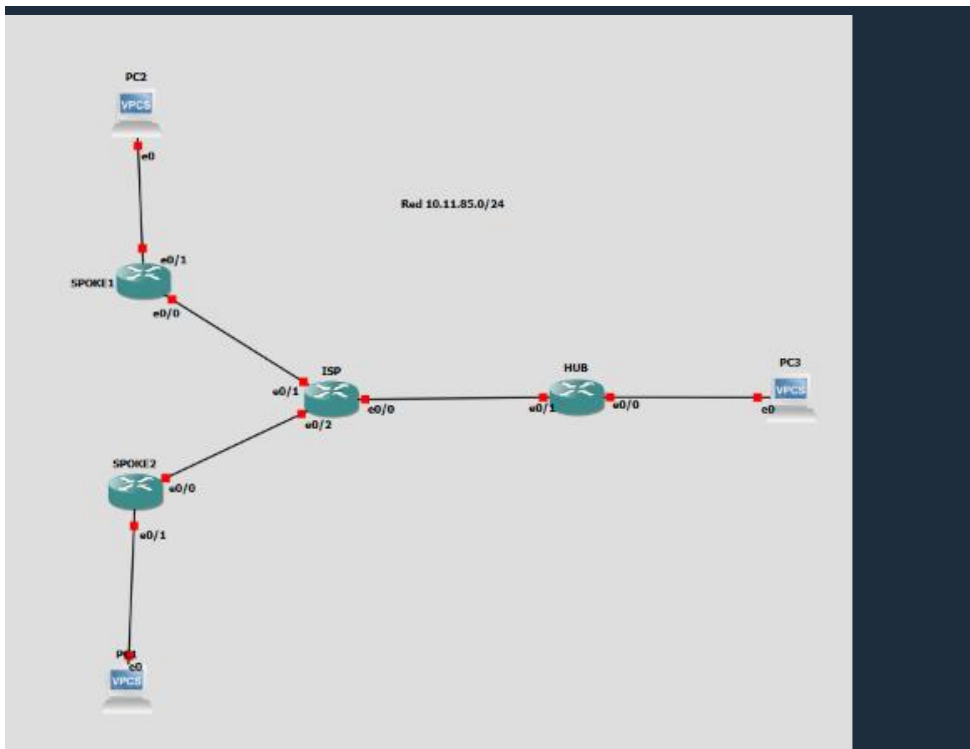


Objetivo

Implementar una VPN hub-and-spoke punto a multipunto DMVPN Fase 2 utilizando IPsec IKEv1 y enrutamiento dinámico EIGRP. DMVPN Fase 2 introduce mGRE (multipoint GRE) y NHRP (Next Hop Resolution Protocol), permitiendo que el HUB administre dinámicamente las direcciones públicas de los spokes sin necesidad de un túnel punto a punto por cada uno. EIGRP propaga las rutas de las LANs entre el HUB y los spokes a través del túnel.

Topología de Red



Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Descripción
HUB	e0/1	10.11.85.98/30	WAN → ISP
HUB	e0/0	10.11.85.65/27	LAN → PC3

SPOKE1	e0/0	10.11.85.102/30	WAN → ISP
SPOKE1	e0/1	10.11.85.1/27	LAN → PC2
SPOKE2	e0/0	10.11.85.106/30	WAN → ISP
SPOKE2	e0/1	10.11.85.33/27	LAN → PC1
ISP	e0/0	10.11.85.97/30	Hacia HUB
ISP	e0/1	10.11.85.101/30	Hacia SPOKE1
ISP	e0/2	10.11.85.105/30	Hacia SPOKE2

Red NBMA del túnel

Dispositivo	IP Tunnel0	Rol
HUB	10.10.10.1/27	NHS (Next Hop Server)
SPOKE1	10.10.10.2/27	Spoke
SPOKE2	10.10.10.3/27	Spoke

Parámetros

IKEv1 (ISAKMP)

Parámetro	Valor
Política	10
Cifrado	AES-256
Hash	SHA-256
Autenticación	PSK (wildcard 0.0.0.0 0.0.0.0)
Grupo DH	14 (2048-bit)
Lifetime	86400 s
Clave PSK	ITLA2026

IPSec / NHRP / Túnel

Parámetro	Valor
Transform Set	TS-DMVPN / ESP / AES-256 / SHA256 / Transport
IPSec Profile	PROF-DMVPN
Tunnel mode	gre multipoint
NHRP network-id	85185
NHRP authentication	ITLA-DMVPN
Tunnel key	85185

EIGRP

Parámetro	Valor
AS Number	1
Auto-summary	Deshabilitado (no auto-summary)
Extra en HUB	no ip split-horizon eigrp 1 + no ip next-hop-self eigrp 1
Extra en Spokes	Ninguno (solo van en el HUB)

Scripts de Configuración

ISP

enable

configure terminal

hostname ISP

interface Ethernet0/0

description Hacia HUB

ip address 10.11.85.97 255.255.255.252

no shutdown

interface Ethernet0/1

description Hacia SPOKE1

ip address 10.11.85.101 255.255.255.252

no shutdown

interface Ethernet0/2

description Hacia SPOKE2

ip address 10.11.85.105 255.255.255.252

no shutdown

ip route 10.11.85.64 255.255.255.224 10.11.85.98

ip route 10.11.85.0 255.255.255.224 10.11.85.102

ip route 10.11.85.32 255.255.255.224 10.11.85.106

end

write memory

HUB

enable

configure terminal

hostname HUB

interface Ethernet0/1

description WAN hacia ISP

ip address 10.11.85.98 255.255.255.252

no shutdown

```
interface Ethernet0/0
description LAN hacia PC3
ip address 10.11.85.65 255.255.255.224
no shutdown
```

```
ip route 0.0.0.0 0.0.0.0 10.11.85.97
crypto isakmp policy 10
encr aes 256
hash sha256
authentication pre-share
group 14
lifetime 86400
```

```
crypto isakmp key ITLA2026 address 0.0.0.0 0.0.0.0
```

```
crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
mode transport
```

```
crypto ipsec profile PROF-DMVPN
set transform-set TS-DMVPN
```

```
interface Tunnel0
description DMVPN Fase2 - Hub
ip address 10.10.10.1 255.255.255.224
no ip redirects
ip mtu 1400
ip nhrp authentication ITLA-DMVPN
ip nhrp map multicast dynamic
ip nhrp network-id 85185
ip nhrp holdtime 300
no ip split-horizon eigrp 1
no ip next-hop-self eigrp 1
tunnel source Ethernet0/1
tunnel mode gre multipoint
tunnel key 85185
tunnel protection ipsec profile PROF-DMVPN
no shutdown
```

```
router eigrp 1
no auto-summary
network 10.10.10.0 0.0.0.31
network 10.11.85.64 0.0.0.31
```

end
write memory

SPOKE1

enable
configure terminal

hostname SPOKE1
interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.102 255.255.255.252
no shutdown

interface Ethernet0/1
description LAN hacia PC2
ip address 10.11.85.1 255.255.255.224
no shutdown
ip route 0.0.0.0 0.0.0.0 10.11.85.101
crypto isakmp policy 10
encr aes 256
hash sha256
authentication pre-share
group 14
lifetime 86400
crypto isakmp key ITLA2026 address 0.0.0.0 0.0.0.0
crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
mode transport
crypto ipsec profile PROF-DMVPN
set transform-set TS-DMVPN

interface Tunnel0
description DMVPN Fase2 - Spoke1
ip address 10.10.10.2 255.255.255.224
no ip redirects
ip mtu 1400
ip nhrp authentication ITLA-DMVPN
ip nhrp map 10.10.10.1 10.11.85.98
ip nhrp map multicast 10.11.85.98
ip nhrp network-id 85185
ip nhrp holdtime 300
ip nhrp nhs 10.10.10.1
tunnel source Ethernet0/0
tunnel mode gre multipoint

```
tunnel key 85185
tunnel protection ipsec profile PROF-DMVPN
no shutdown
router eigrp 1
no auto-summary
network 10.10.10.0 0.0.0.31
network 10.11.85.0 0.0.0.31
end
write memory
```

SPOKE2

```
enable
configure terminal
```

```
hostname SPOKE2
interface Ethernet0/0
description WAN hacia ISP
ip address 10.11.85.106 255.255.255.252
no shutdown
```

```
interface Ethernet0/1
description LAN hacia PC1
ip address 10.11.85.33 255.255.255.224
no shutdown
ip route 0.0.0.0 0.0.0.0 10.11.85.105
```

```
crypto isakmp policy 10
encr aes 256
hash sha256
authentication pre-share
group 14
lifetime 86400
```

```
crypto isakmp key ITLA2026 address 0.0.0.0 0.0.0.0
crypto ipsec transform-set TS-DMVPN esp-aes 256 esp-sha256-hmac
mode transport
crypto ipsec profile PROF-DMVPN
set transform-set TS-DMVPN
interface Tunnel0
description DMVPN Fase2 - Spoke2
ip address 10.10.10.3 255.255.255.224
no ip redirects
ip mtu 1400
ip nhrp authentication ITLA-DMVPN
```

```
ip nhrp map 10.10.10.1 10.11.85.98
ip nhrp map multicast 10.11.85.98
ip nhrp network-id 85185
ip nhrp holdtime 300
ip nhrp nhs 10.10.10.1
tunnel source Ethernet0/0
tunnel mode gre multipoint
tunnel key 85185
tunnel protection ipsec profile PROF-DMVPN
no shutdown
```

```
router eigrp 1
no auto-summary
network 10.10.10.0 0.0.0.31
network 10.11.85.32 0.0.0.31
```

```
end
write memory
```

VPCS

PC1

```
ip 10.11.85.34 255.255.255.224 10.11.85.33
```

save

PC2

```
ip 10.11.85.2 255.255.255.224 10.11.85.1
```

save

PC3

```
ip 10.11.85.66 255.255.255.224 10.11.85.65
```

save

Verificación

! 1. En ISP: confirmar rutas estáticas show ip route static

```
ISP#show ip route static
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override
```

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks

```
S      10.11.85.0/27 [1/0] via 10.11.85.102
S      10.11.85.32/27 [1/0] via 10.11.85.106
S      10.11.85.64/27 [1/0] via 10.11.85.98
red#
```

! 2. IKE Fase 1

```
show crypto isakmp sa
```

```
HUB#show crypto isakmp sa
```

```
IPv4 Crypto ISAKMP SA
```

dst	src	state	conn-id	status
10.11.85.98	10.11.85.102	QM_IDLE	1001	ACTIVE
10.11.85.98	10.11.85.106	QM_IDLE	1002	ACTIVE

```
IPv6 Crypto ISAKMP SA
```

```

SPOKE1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA

```

dst	src	state	conn-id	status
10.11.85.106	10.11.85.102	QM_IDLE	1002	ACTIVE
10.11.85.98	10.11.85.102	QM_IDLE	1001	ACTIVE

```

IPv6 Crypto ISAKMP SA
SPOKE1#show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 10.11.85.102

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.11.85.102/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (10.11.85.106/255.255.255.255/47/0)
  current_peer 10.11.85.106 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 5, #pkts encrypt: 5, #pkts digest: 5
    #pkts decaps: 4, #pkts decrypt: 4, #pkts verify: 4
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

  local crypto endpt.: 10.11.85.102, remote crypto endpt.: 10.11.85.106
  plaintext mtu 1458, path mtu 1500, ip mtu 1500, ip mtu idb (none)
  current outbound spi: 0x28464854(675694676)
  PFS (Y/N): N, DH group: none

  inbound esp sas:
    spi: 0x900A6D14(2416602388)

```

! 3. Registro NHRP

show ip nhrp

```

HUB#show ip nhrp
10.10.10.2/32 via 10.10.10.2
  Tunnel0 created 01:24:05, expire 00:04:05
  Type: dynamic, Flags: unique registered used nhop
  NBMA address: 10.11.85.102
10.10.10.3/32 via 10.10.10.3
  Tunnel0 created 01:23:47, expire 00:04:23
  Type: dynamic, Flags: unique registered used nhop
  NBMA address: 10.11.85.106
HUB#

```

! 4. Estado de la nube DMVPN

show dmvpn

```

HUB#
HUB#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:2,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
      1 10.11.85.102      10.10.10.2    UP 01:24:17    D
      1 10.11.85.106      10.10.10.3    UP 01:23:59    D

HUB#

```

```

SPOKE2#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
        N - NATed, L - Local, X - No Socket
        # Ent --> Number of NHRP entries with same NBMA peer
        NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
        UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Spoke, NHRP Peers:1,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
      1 10.11.85.98      10.10.10.1    UP 01:25:33    S

```

! 5. Estado del túnel
show interface Tunnel0

```

HUB#show interface Tunnel0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Description: DMVPN Fase2 - Hub
  Internet address is 10.10.10.1/27
  MTU 17912 bytes, BW 100 Kbit/sec, DLY 50000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 10.11.85.98 (Ethernet0/1)
  Tunnel Subblocks:
    src-track:
      Tunnel0 source tracking subblock associated with Ethernet0/1
      Set of tunnels with source Ethernet0/1, 1 member (includes iterators), on interface <OK>
  Tunnel protocol/transport multi-GRE/IP
  Key 0x14CC1, sequencing disabled
  Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1472 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Tunnel protection via IPSec (profile "PROF-DMVPN")
  Last input 00:00:03, output never, output hang never
  Last clearing of "show interface" counters 01:25:03
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    2309 packets input, 206691 bytes, 0 no buffer
    Received 0 broadcasts (0 IP multicasts)
    0 runs, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    2314 packets output, 209583 bytes, 0 underruns
    0 output errors, 0 collisions, 0 interface resets
    0 unknown protocol drops
    0 output buffer failures, 0 output buffers swapped out
HUB#

```

! 6. Vecinos EIGRP

```

show ip eigrp neighbors
0 output buffer failures, 0 output buffers swapped out

```

```

HUB# show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(1)

```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num
1	10.10.10.3	Tu0	13	01:24:29	17	1440	0	3
0	10.10.10.2	Tu0	13	01:24:43	14	1440	0	4

```

HUB#

```

```

HUB#

```

! 7. Rutas EIGRP aprendidas

```

show ip route eigrp

```

```
HUB#show ip route eigrp
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override
```

```
Gateway of last resort is 10.11.85.97 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
```

```
D 10.11.85.0/27 [90/26905600] via 10.10.10.2, 01:24:56, Tunnel0
```

```
D 10.11.85.32/27 [90/26905600] via 10.10.10.3, 01:24:41, Tunnel0
```

```
SPOKE2#show ip route eigrp
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override
```

```
Gateway of last resort is 10.11.85.105 to network 0.0.0.0
```

```
10.0.0.0/8 is variably subnetted, 8 subnets, 3 masks
```

```
D 10.11.85.0/27 [90/28185600] via 10.10.10.2, 01:25:01, Tunnel0
```

```
D 10.11.85.64/27 [90/26905600] via 10.10.10.1, 01:25:01, Tunnel0
```